

ASSIGNMENT-I QUESTIONS

COURSE CODE: CSE 103

COURSE NAME: DISCRETE MATHEMATICS

SOURCE: Book- Discrete Mathematics and Its Applications by Kenneth H. Rosen (7th Edition)

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14. Let p , q , and r be the propositions

p : You get an A on the final exam.

q : You do every exercise in this book.

r : You get an A in this class.

Write these propositions using p , q , and r and logical connectives (including negations).

a) You get an A in this class, but you do not do every exercise in this book.

b) You get an A on the final, you do every exercise in this book, and you get an A in this class.

c) To get an A in this class, it is necessary for you to get an A on the final.

d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.

e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.

f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

15. Let p , q , and r be the propositions

p : Grizzly bears have been seen in the area.

q : Hiking is safe on the trail.

r : Berries are ripe along the trail.

Write these propositions using p , q , and r and logical connectives (including negations).

a) Berries are ripe along the trail, but grizzly bears have not been seen in the area.

b) Grizzly bears have not been seen in the area and hiking on the trail is safe, but berries are ripe along the trail.

c) If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.

d) It is not safe to hike on the trail, but grizzly bears have not been seen in the area and the berries along the trail are ripe.

e) For hiking on the trail to be safe, it is necessary but not sufficient that berries not be ripe along the trail and for grizzly bears not to have been seen in the area.

f) Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.

23. Write each of these statements in the form “if p, then q” in English. [Hint: Refer to the list of common ways to express conditional statements.]

- a) It snows whenever the wind blows from the northeast.
- b) The apple trees will bloom if it stays warm for a week.
- c) That the Pistons win the championship implies that they beat the Lakers.
- d) It is necessary to walk 8 miles to get to the top of Long’s Peak.
- e) To get tenure as a professor, it is sufficient to be world-famous.
- f) If you drive more than 400 miles, you will need to buy gasoline.
- g) Your guarantee is good only if you bought your CD player less than 90 days ago.
- h) Jan will go swimming unless the water is too cold.

24. Write each of these statements in the form “if p, then q” in English. [Hint: Refer to the list of common ways to express conditional statements provided in this section.]

- a) I will remember to send you the address only if you send me an e-mail message.
- b) To be a citizen of this country, it is sufficient that you were born in the United States.
- c) If you keep your textbook, it will be a useful reference in your future courses.
- d) The Red Wings will win the Stanley Cup if their goalie plays well.
- e) That you get the job implies that you had the best credentials.
- f) The beach erodes whenever there is a storm.
- g) It is necessary to have a valid password to log on to the server.
- h) You will reach the summit unless you begin your climb too late.

27. State the converse, contrapositive, and inverse of each of these conditional statements.

- a) If it snows today, I will ski tomorrow.
- b) I come to class whenever there is going to be a quiz.
- c) A positive integer is a prime only if it has no divisors other than 1 and itself.

28. State the converse, contrapositive, and inverse of each of these conditional statements.

- a) If it snows tonight, then I will stay at home.
- b) I go to the beach whenever it is a sunny summer day.
- c) When I stay up late, it is necessary that I sleep until noon.

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7. Express these system specifications using the propositions p “The message is scanned for viruses” and q “The message was sent from an unknown system” together with logical connectives (including negations).

- a) “The message is scanned for viruses whenever the message was sent from an unknown system.”
- b) “The message was sent from an unknown system but it was not scanned for viruses.”
- c) “It is necessary to scan the message for viruses whenever it was sent from an unknown system.”
- d) “When a message is not sent from an unknown system it is not scanned for viruses.”

8. Express these system specifications using the propositions p “The user enters a valid password,” q “Access is granted,” and r “The user has paid the subscription fee” and logical connectives (including negations).

- a) “The user has paid the subscription fee, but does not enter a valid password.”
- b) “Access is granted whenever the user has paid the subscription fee and enters a valid password.”
- c) “Access is denied if the user has not paid the subscription fee.”
- d) “If the user has not entered a valid password but has paid the subscription fee, then access is granted.”

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10. Show that each of these conditional statements is a tautology by using truth tables.

- a) $[\neg p \wedge (p \vee q)] \rightarrow q$
- b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
- c) $[p \wedge (p \rightarrow q)] \rightarrow q$
- d) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

Answer:

- a) $[\neg p \wedge (p \vee q)] \rightarrow q$
- b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
- c) $[p \wedge (p \rightarrow q)] \rightarrow q$
- d) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

30. Show that $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology.

31. Show that $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ are not logically equivalent.

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24. Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.

- a) Everyone in your class has a cellular phone.
- b) Somebody in your class has seen a foreign movie.
- c) There is a person in your class who cannot swim.
- d) All students in your class can solve quadratic equations.
- e) Some student in your class does not want to be rich.

27. Translate each of these statements into logical expressions in three different ways by varying the domain and by using predicates with one and with two variables.

- a) A student in your school has lived in Vietnam.
- b) There is a student in your school who cannot speak Hindi.
- c) A student in your school knows Java, Prolog, and C++.
- d) Everyone in your class enjoys Thai food.
- e) Someone in your class does not play hockey.

30. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y , where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.

- a) $\exists x P(x, 3)$
- b) $\forall y P(1, y)$
- c) $\exists y \neg P(2, y)$
- d) $\forall x \neg P(x, 2)$

39. Translate these specifications into English where $F(p)$ is “Printer p is out of service,” $B(p)$ is “Printer p is busy,” $L(j)$ is “Print job j is lost,” and $Q(j)$ is “Print job j is queued.”

- a) $\exists p(F(p) \wedge B(p)) \rightarrow \exists j L(j)$
- b) $\forall p B(p) \rightarrow \exists j Q(j)$

$$c) \exists j(Q(j) \wedge L(j)) \rightarrow \exists p F(p)$$

$$d) (\forall p B(p) \wedge \forall j Q(j)) \rightarrow \exists j L(j)$$

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7. Let $T(x, y)$ mean that student x likes cuisine y , where the domain for x consists of all students at your school and the domain for y consists of all cuisines. Express each of these statements by a simple English sentence.

$$a) \neg T(\text{Abdallah Hussein}, \text{Japanese})$$

$$b) \exists x T(x, \text{Korean}) \wedge \forall x T(x, \text{Mexican})$$

$$c) \exists y(T(\text{Monique Arsenault}, y) \vee T(\text{Jay Johnson}, y))$$

$$d) \forall x \forall z \exists y((x = z) \rightarrow \neg(T(x, y) \wedge T(z, y)))$$

$$e) \exists x \exists z \forall y(T(x, y) \leftrightarrow T(z, y))$$

$$f) \forall x \forall z \exists y(T(x, y) \leftrightarrow T(z, y))$$

8. Let $Q(x, y)$ be the statement “student x has been a contestant on quiz show y .” Express each of these sentences in terms of $Q(x, y)$, quantifiers, and logical connectives, where the domain for x consists of all students at your school and for y consists of all quiz shows on television.

a) There is a student at your school who has been a contestant on a television quiz show.

b) No student at your school has ever been a contestant a television quiz show.

c) There is a student at your school who has been a contestant on Jeopardy and on Wheel of Fortune.

d) Every television quiz show has had a student from your school as a contestant.

e) At least two students from your school have been contestants on Jeopardy.

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5. Use rules of inference to show that the hypotheses “Randy works hard,” “If Randy works hard, then he is a dull boy,” and “If Randy is a dull boy, then he will not get the job” imply the conclusion “Randy will not get the job.”

6. Use rules of inference to show that the hypotheses “If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on,” “If the sailing race is held, then the trophy will be awarded,” and “The trophy was not awarded” imply the conclusion “It rained.”

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10. Determine whether each of these functions from $\{a, b, c, d\}$ to itself is one-to-one.

a) $f(a) = b, f(b) = a, f(c) = c, f(d) = d$

b) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$

c) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$

12. Determine whether each of these functions from $\mathbf{Z}$ to $\mathbf{Z}$ is one-to-one.

a) $f(n) = n - 1$

b) $f(n) = n^2 + 1$

c) $f(n) = n^3$

d) $f(n) = \frac{n}{2}$

13. Which functions in Exercise 12 are onto?

23. Determine whether each of these functions is a bijection from $\mathbf{R}$ to $\mathbf{R}$.

a) $f(x) = 2x + 1$

b) $f(x) = x^2 + 1$

c) $f(x) = x^3$

d) $f(x) = \frac{(x^2+1)}{(x^2+2)}$